

Project Presentation

Myocardial Infarction Severity stages classification from ECG signals using ST-GNN

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Introduction

Myocardial infarction, commonly called a heart attack, is a critical cardiac condition that requires rapid and accurate diagnosis to reduce complications and mortality. Although ECG is one of the most widely used diagnostic tools, manual interpretation can be difficult because ECG patterns vary across patients, leads, and infarction severity levels.

This project focuses on automated myocardial infarction detection and severity analysis using 12-lead ECG signals with a Spatial-Temporal Graph Neural Network architecture.

The work uses ECG records from PTB and STAFF III datasets, applies preprocessing such as band-pass filtering, wavelet denoising, LMS adaptive filtering, segmentation, and downsampling, and then combines waveform data with extracted clinical features for classification.

Importance and Relevance

- Early ECG-based detection can improve diagnosis speed and support timely treatment.
- Advanced deep learning models can learn complex spatial and temporal ECG patterns more effectively than traditional approaches.
- This work has real-world relevance in emergency care, hospitals, and decision-support systems for cardiac screening.
- The inclusion of clinical features like QRS width, heart-rate proxy, and ST deviation makes the approach more practical for real clinical applications.

Problem Statement

- Detecting myocardial infarction from ECG signals is difficult because ECG patterns change across patients and leads.
- Manual analysis is slow and can be inconsistent.
- ECG data contains noise and requires preprocessing such as filtering, denoising, segmentation, and feature extraction.
- Existing methods may not fully capture both multi-lead relationships and time-based ECG changes.
- So, a more advanced model is needed for accurate and reliable prediction.

Objectives

Main goal: To develop an intelligent system for myocardial infarction detection using 12-lead ECG signals and an ST-GNN model.

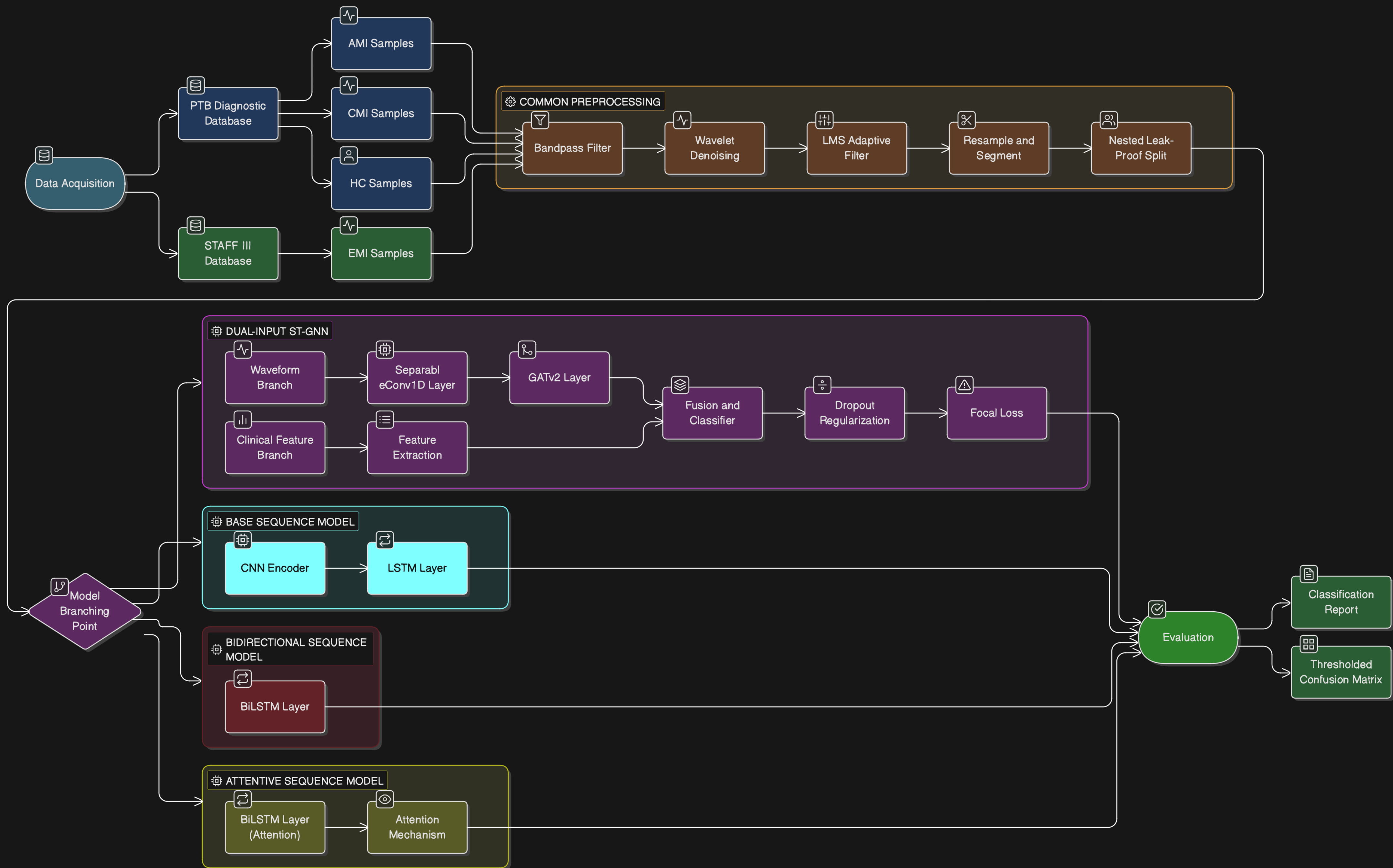
- To preprocess ECG signals using filtering, denoising, segmentation, and downsampling.
- To extract important clinical features such as QRS width, heart-rate proxy, and ST deviation.
- To build a model that combines waveform data and clinical features for accurate classification.

Literature Review

- Traditional methods used manual ECG feature extraction and machine learning classifiers.
- Deep learning methods now learn ECG features automatically from data.
- Common models include CNN, LSTM, CRNN, GRU, and ResNet.
- Most studies use 12-lead ECG datasets such as PTB and PTB-XL.
- Preprocessing often includes denoising, filtering, segmentation, and normalization.
- Some models detect MI, while others also localize the infarction area.
- Research gaps include noise sensitivity, poor real-world generalization, and difficulty handling complex cases

Methodology

- Collect multi-lead ECG signals from the patient using standard electrodes and lead wires.
- Amplify, filter, and convert the raw ECG voltages into clean digital samples.
- Transfer digitized ECG data from the front-end hardware to the processing device
- Apply digital filtering, denoising, resampling, and segmentation to obtain fixed-length ECG windows.
- Organize ECG segments, assign diagnostic labels, balance classes, and split into train/validation/test sets.
- Use ML/DL models to learn features and classify each ECG segment.
- Train the models, tune hyper parameters, and select the best configuration based on validation performance.



Technology used

- Programming Language: Python.
- Libraries: NumPy, Pandas, SciPy, PyWavelets, WFDB, TensorFlow, Keras, Scikit-learn, and Matplotlib.
- Deep Learning Tools: TensorFlow/Keras with layers such as SeparableConv1D, BatchNormalization, Dense, Dropout, and GlobalAveragePooling1D.
- Training Tools: Adam optimizer, EarlyStopping, and ReduceLROnPlateau.
- Dataset Support: PTB and STAFF III ECG databases processed using WFDB.
- Hardware: 12th Gen Intel(R) Core(TM) i7-12700 (2.10 GHz), 500GB SSD, 16GB RAM, 4GB Graphics Card

Database

PTB & STAFF III

1. PTB Diagnostic ECG:

- 549 high-res 12-lead recordings, 1000 Hz
- AMI/CMI/HC + clinical metadata (MI stage/localization).

2. STAFF III Database:

- 104 patients, 12-lead ECGs, 1000 Hz
- Induced ischemia (EMI) + diverse demographics.
- Complements PTB coverage.

• Combined Usage:

- 4-class: HC | EMI | CMI | AMI
- Leak-proof patient splits

Preprocessing

1. Bandpass Filtering

- Purpose: Remove baseline wander (<0.5 Hz) and high-frequency noise (>40 Hz)
- Transfer Function: $H(f) = H_{\text{high}}(f) \times H_{\text{low}}(f)$
- Result: Preserves QRS complex and ST-segment features

$$H(f) = \frac{1}{1 + \left(\frac{f_{\text{low}}}{f}\right)^{2N} + \left(\frac{f}{f_{\text{high}}}\right)^{2N}}$$

2. Wavelet Denoising (Symlet-4)

- DWT decomposes ECG into multi-resolution coefficients.
- Soft thresholding on detail coeffs using universal threshold estimate.
- Benefit: Removes noise while preserving sharp QRS transitions.
- Preserves sharp ECG transients (QRS edges).
- Perfect for non-stationary MI signals.

3. LMS Adaptive Filtering

- Weight Update: $w(n+1) = w(n) + \mu \cdot e(n) \cdot x(n)$
- Error Signal: $e(n) = d(n) - y(n)$
where $d(n)$ = original, $y(n)$ = filtered
- Application: Uses 1-step rolled signal as reference for artifact removal.

4. Resample & Segment

- Downsampling: 1000 Hz → 125 Hz (reduces computation, retains clinical features)
- Segmentation: 4-second windows → 500 samples per segment

Proposed Model

Spatio-Temporal GNN (ST-GNN)

Core Concept

Transform 12-lead ECG into spatial graph where:

- Nodes = Individual ECG leads (I, II, V1-V6, aVR-aVF...)
- Edges = Anatomical adjacency between leads

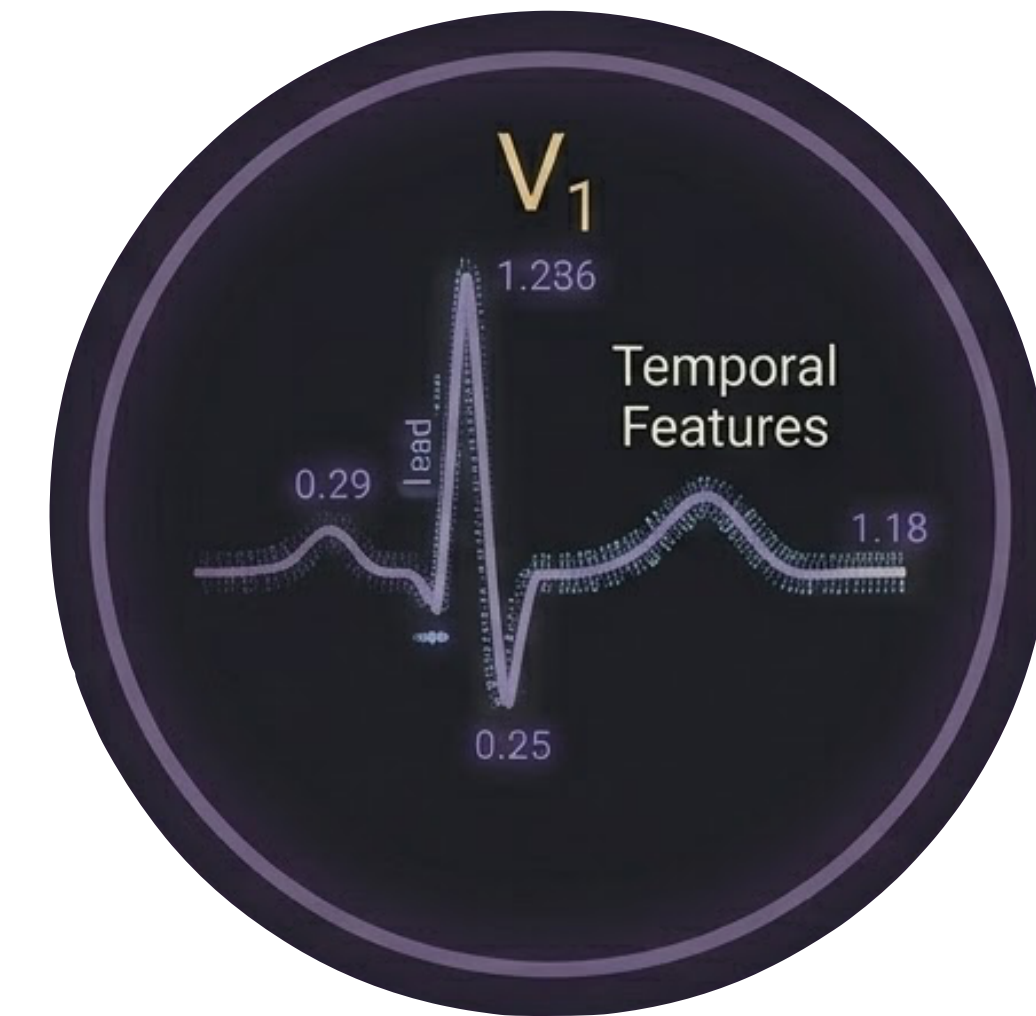
Dual Branch Design

Waveform Branch (Raw Signal Processing)

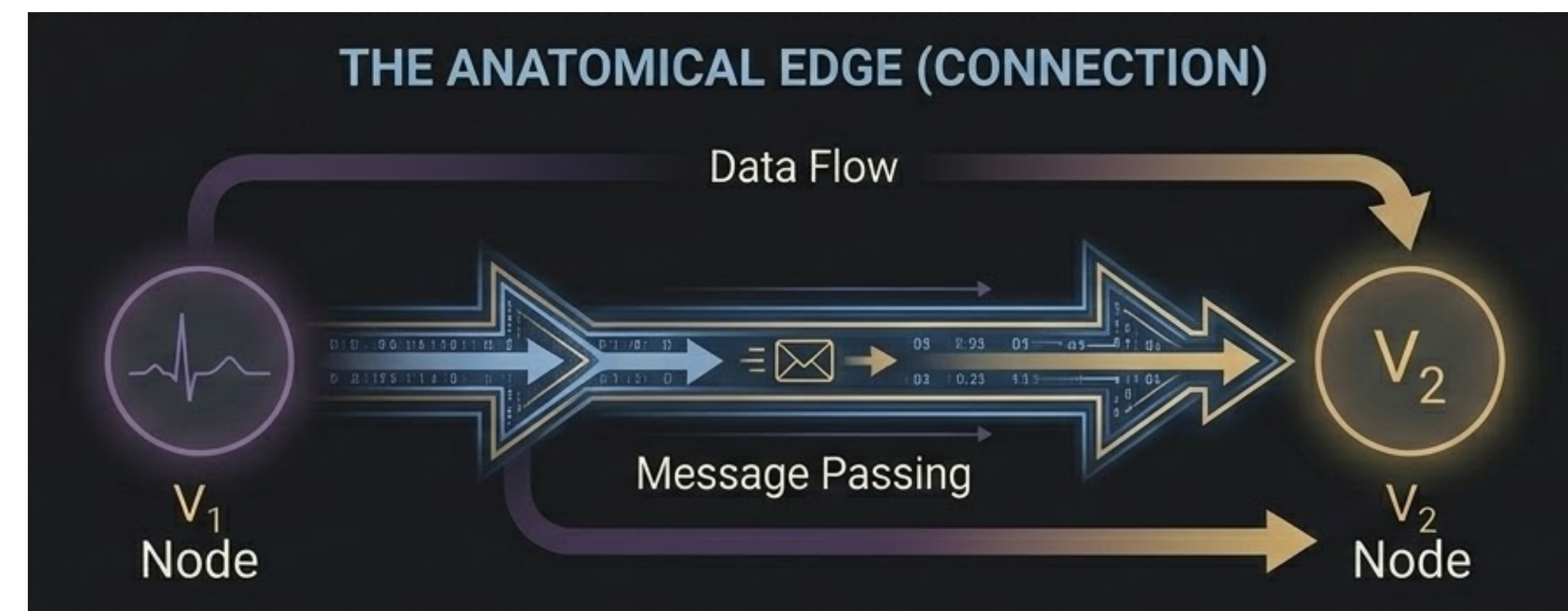
- SeparableConv1D → Temporal feature extraction per lead
- GATv2 Graph Layers → Spatial attention across leads

Clinical Branch (Hand-crafted Features)

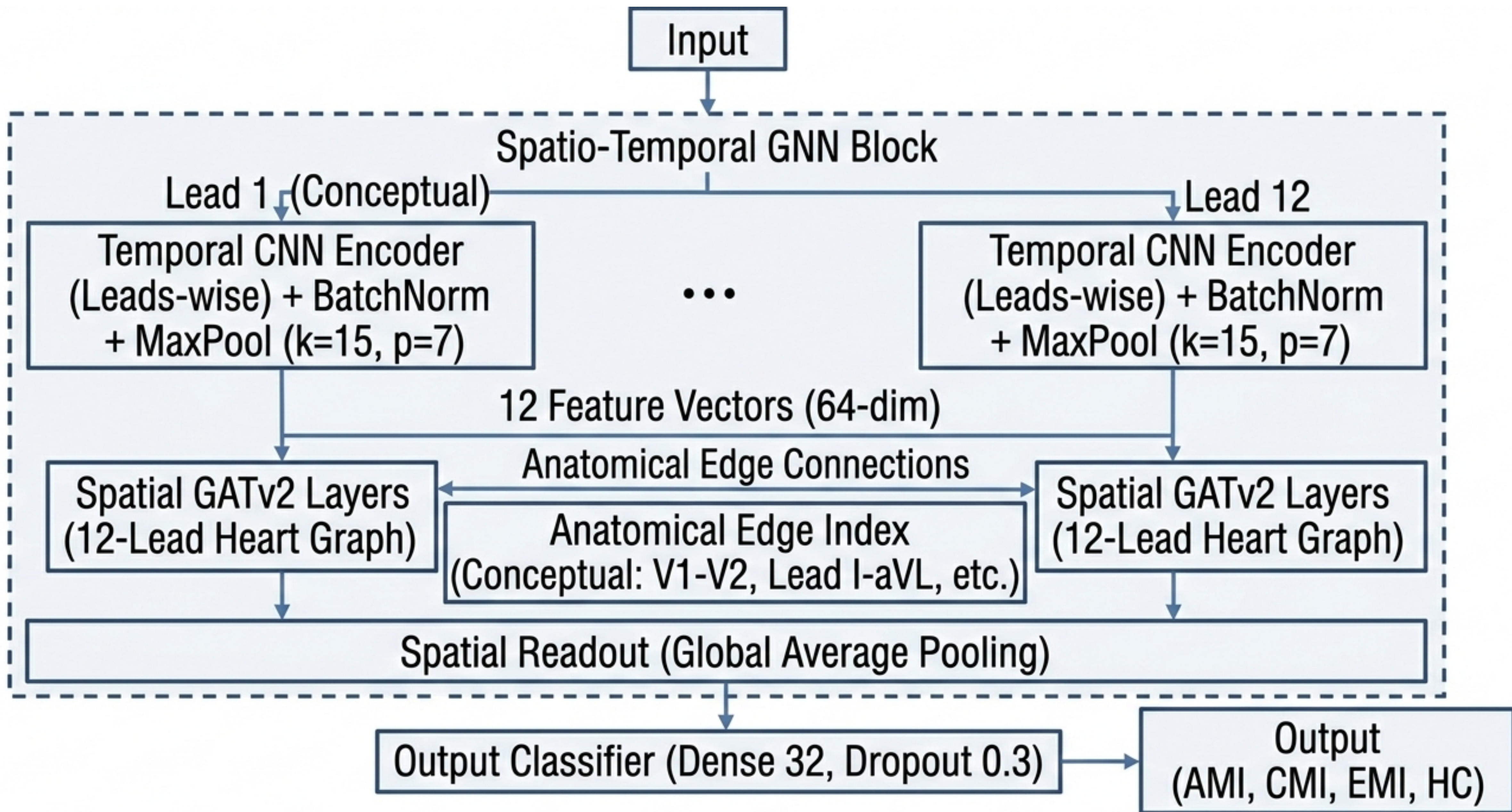
- 36 ECG metrics (QRS duration, ST deviation, HR proxies)
- Dense layers → Interpretable embeddings



NODE



EDGE



ST-GNN

GAT V2

A Graph Attention Network uses an attention mechanism to assign different “weights” (importance scores) to different neighbors. Instead of treating all connected nodes equally, it learns which neighbors are most relevant.

Nodes: The 12 individual ECG leads. The features of each node are the temporal embeddings extracted by the CNN.

Edges: Dictated by your anatomical adjacency matrix. Leads that monitor the same cardiac region (eg: V1-V2) are connected.

- **Diagnostic Model Head**

- Structure: Dropout (0.3) → Dense (32, ReLU) → Batch Norm → Output (4 classes)
- Purpose: Designed to prevent the model from simply memorizing the training data. The single 32-unit Dense layer is lightweight and efficient, relying heavily on the spatial-temporal features extracted earlier.

- **Clinical Model Head**

- Structure: Dense (128, ReLU) → Batch Norm → Dropout (0.4) → Dense (64, ReLU) → Output (4 classes)
- Purpose: Built to handle the combined complexity of raw ECG waveforms merged with engineered clinical features (like QRS width and heart rate proxy)

ST-GNN

Fusion, Regularization & Loss

1. Feature Fusion & Classifier

- Concatenation: Merges ST-GNN waveform features with 36 clinical metrics (QRS width, HR, ST-Deviation) into a single tensor.
- Deep Classifier: The combined data routes through a 128 → 64 unit dense network, ending in a 4-class Softmax output.

2. Dropout Regularization

- Diagnostic Model (0.3): Lighter dropout for the simpler 32-unit head to prevent basic data memorization.
- Clinical Model (0.4): Stronger dropout for the deeper 128-unit head, preventing overfitting on the complex, fused multimodal features.

3. Focal Loss

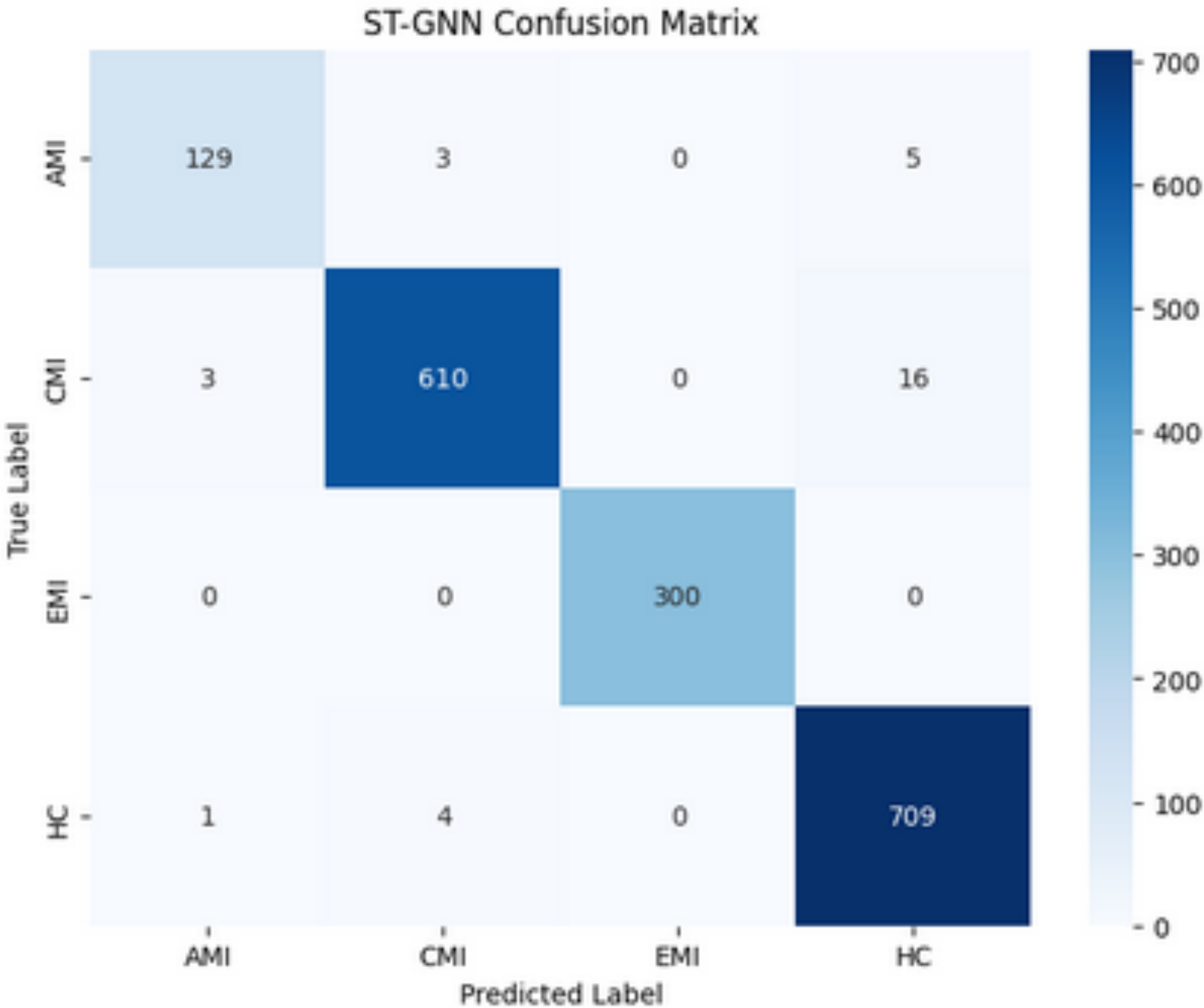
- Handling Imbalance: Replaces standard cross-entropy to address uneven data distribution across MI severity classes.
- Focusing Mechanism: The γ parameter down-weights easy majority-class predictions, forcing the model to prioritize learning hard-to-predict, minority-class cases.

Results

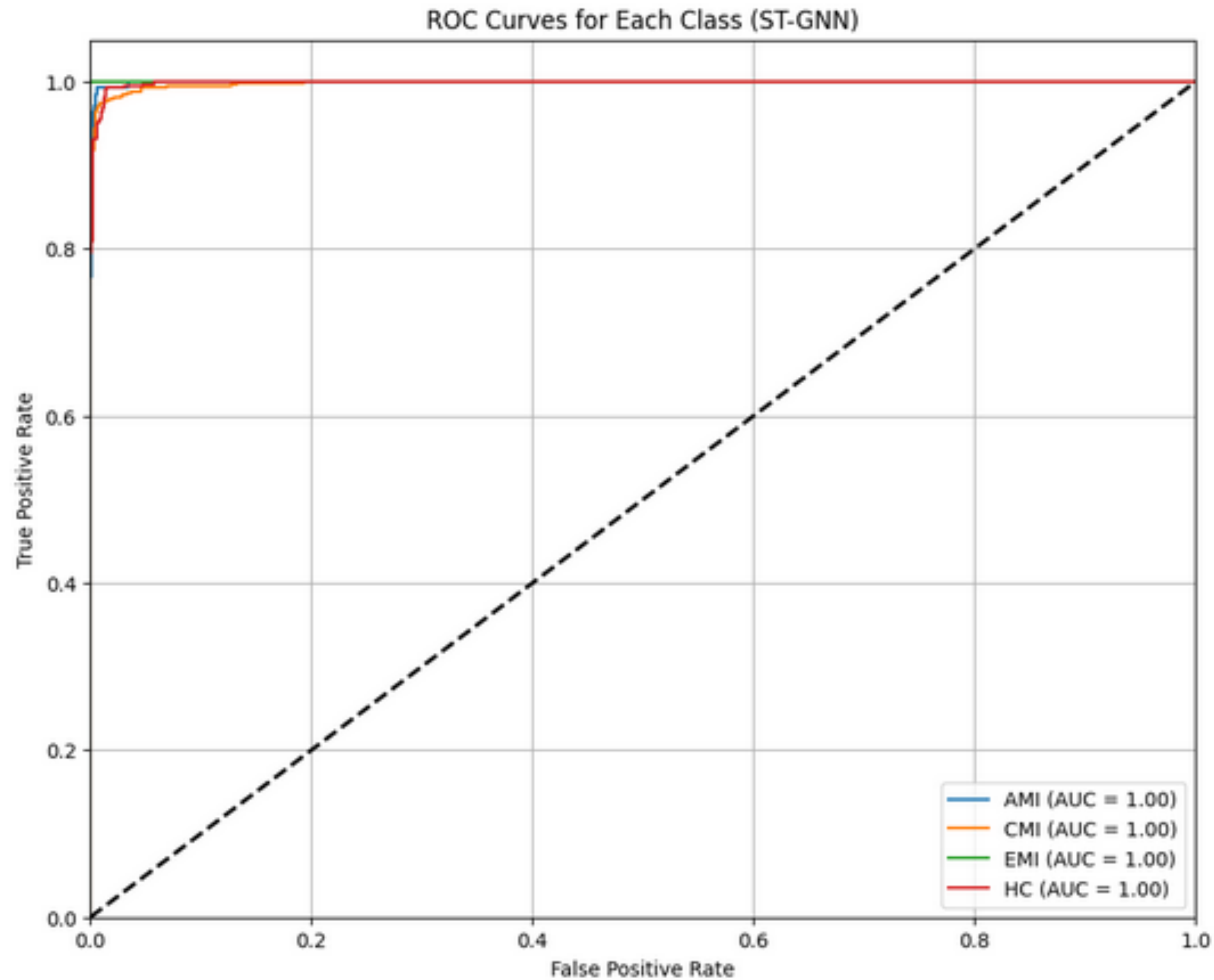
Classification Report of ST-GNN architecture

Classification Report:

	precision	recall	f1-score	support
AMI	0.97	0.94	0.96	137
CMI	0.99	0.97	0.98	629
EMI	1.00	1.00	1.00	300
HC	0.97	0.99	0.98	714
accuracy			0.98	1780
macro avg	0.98	0.98	0.98	1780
weighted avg	0.98	0.98	0.98	1780

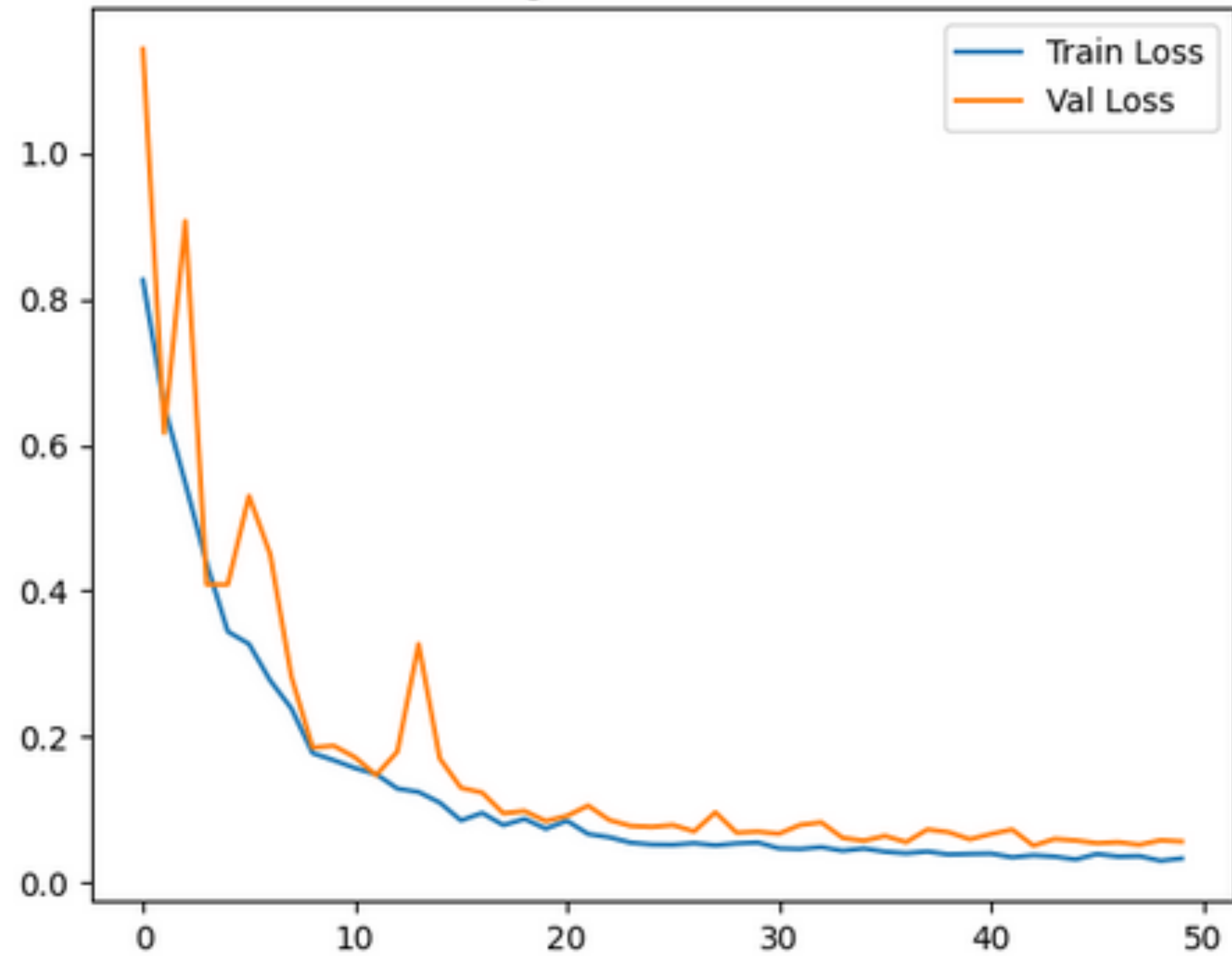


ROC curve of ST-GNN architecture

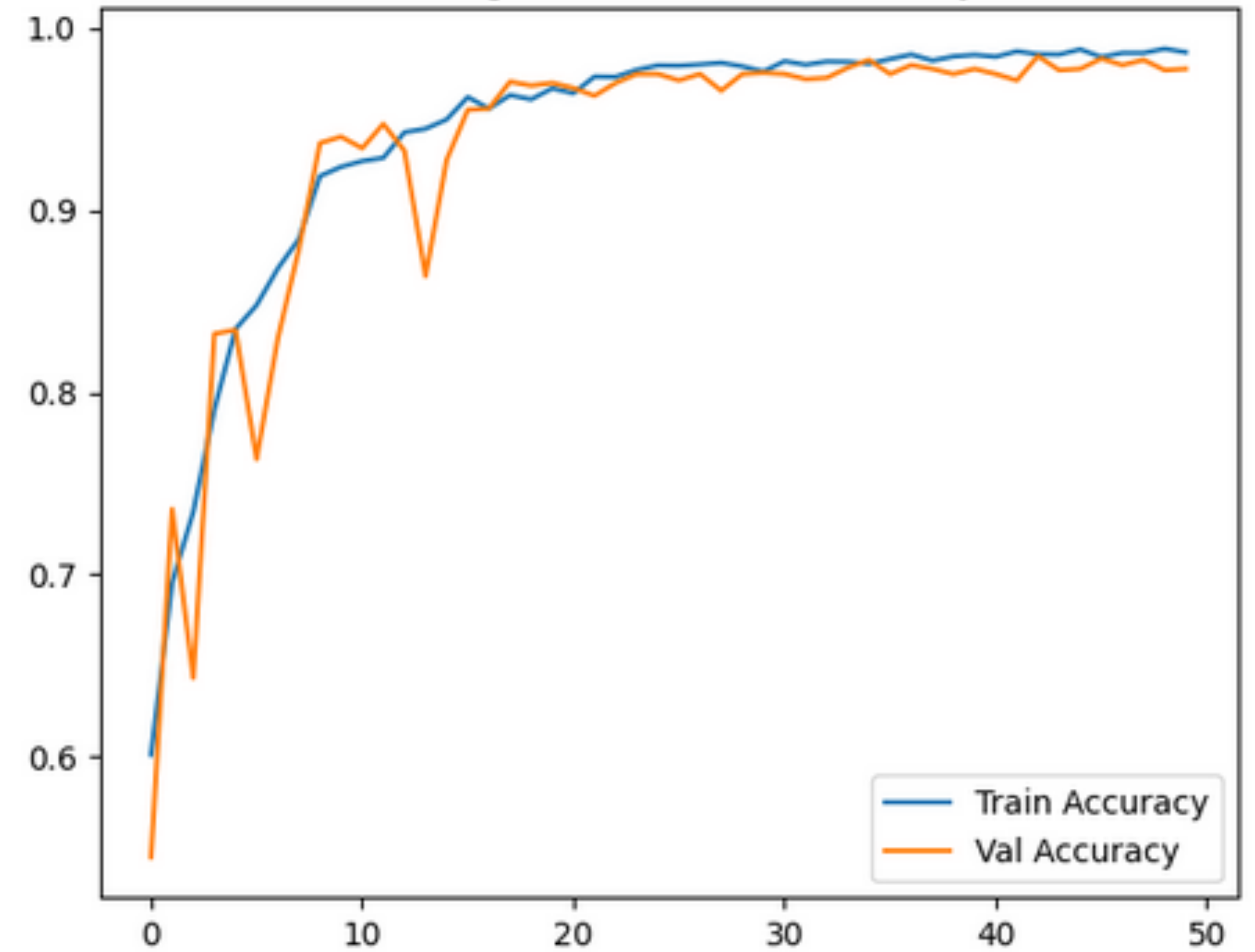


Training vs Validation of ST-GNN architecture

Training and Validation Loss



Training and Validation Accuracy



Classification Report of CNN-LSTM architecture

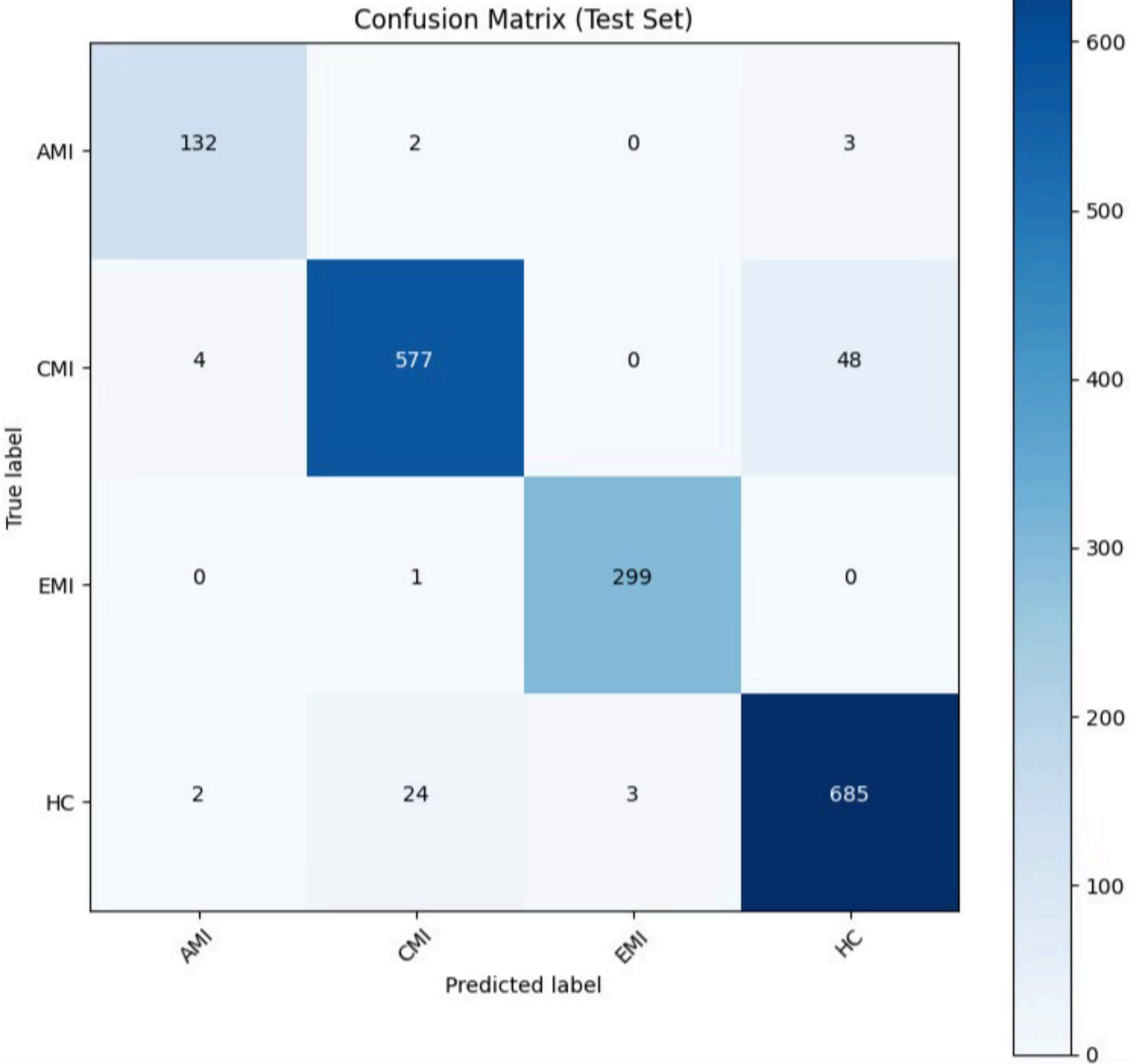
--- Final Results ---

Test Accuracy: 0.9510

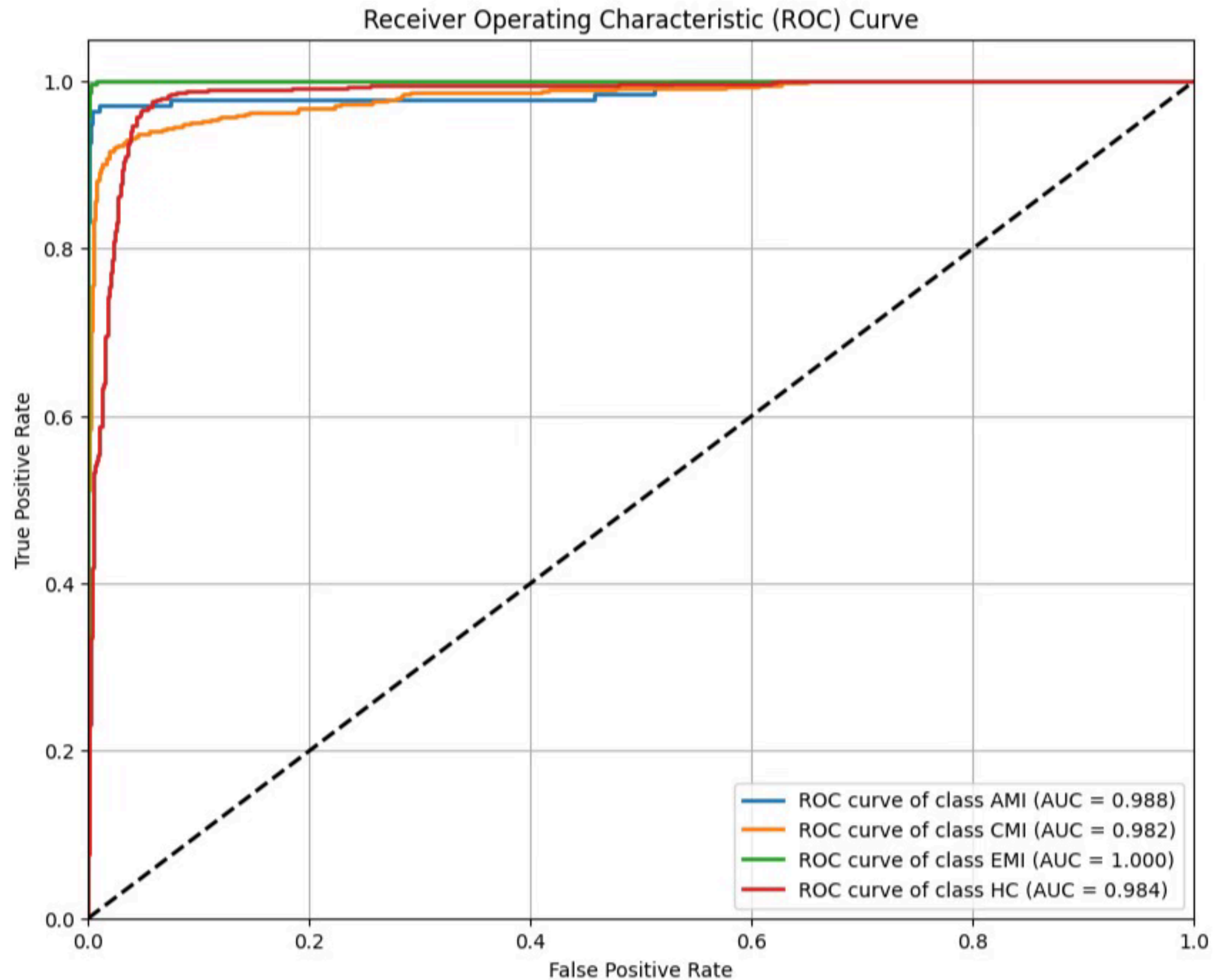
41/41 12s 267ms/step

Classification Report:

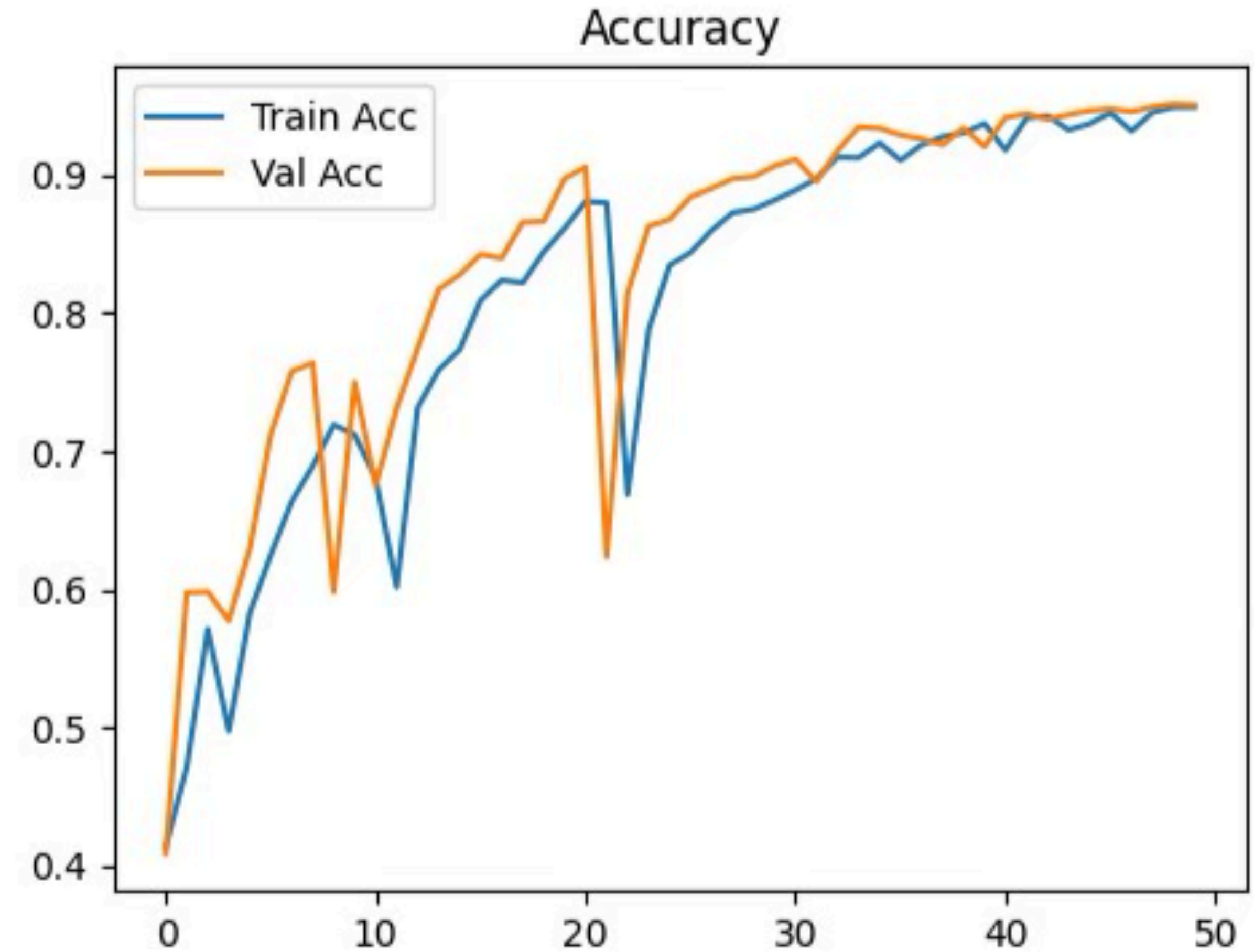
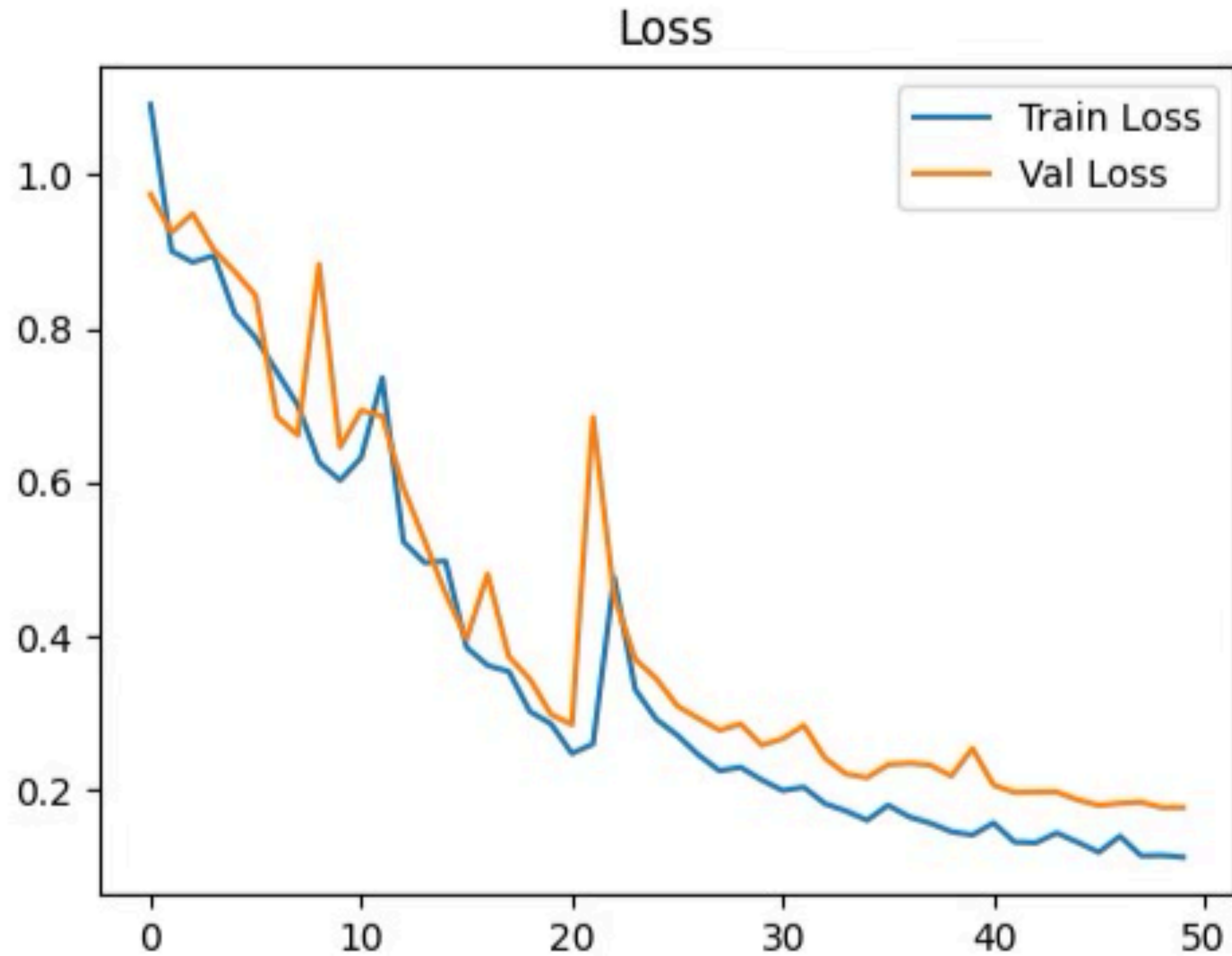
	precision	recall	f1-score	support
AMI	0.95	0.95	0.95	108
CMI	0.95	0.91	0.93	449
EMI	1.00	1.00	1.00	300
HC	0.92	0.96	0.94	429
accuracy			0.95	1286
macro avg	0.96	0.96	0.96	1286
weighted avg	0.95	0.95	0.95	1286



ROC curve of CNN-LSTM architecture



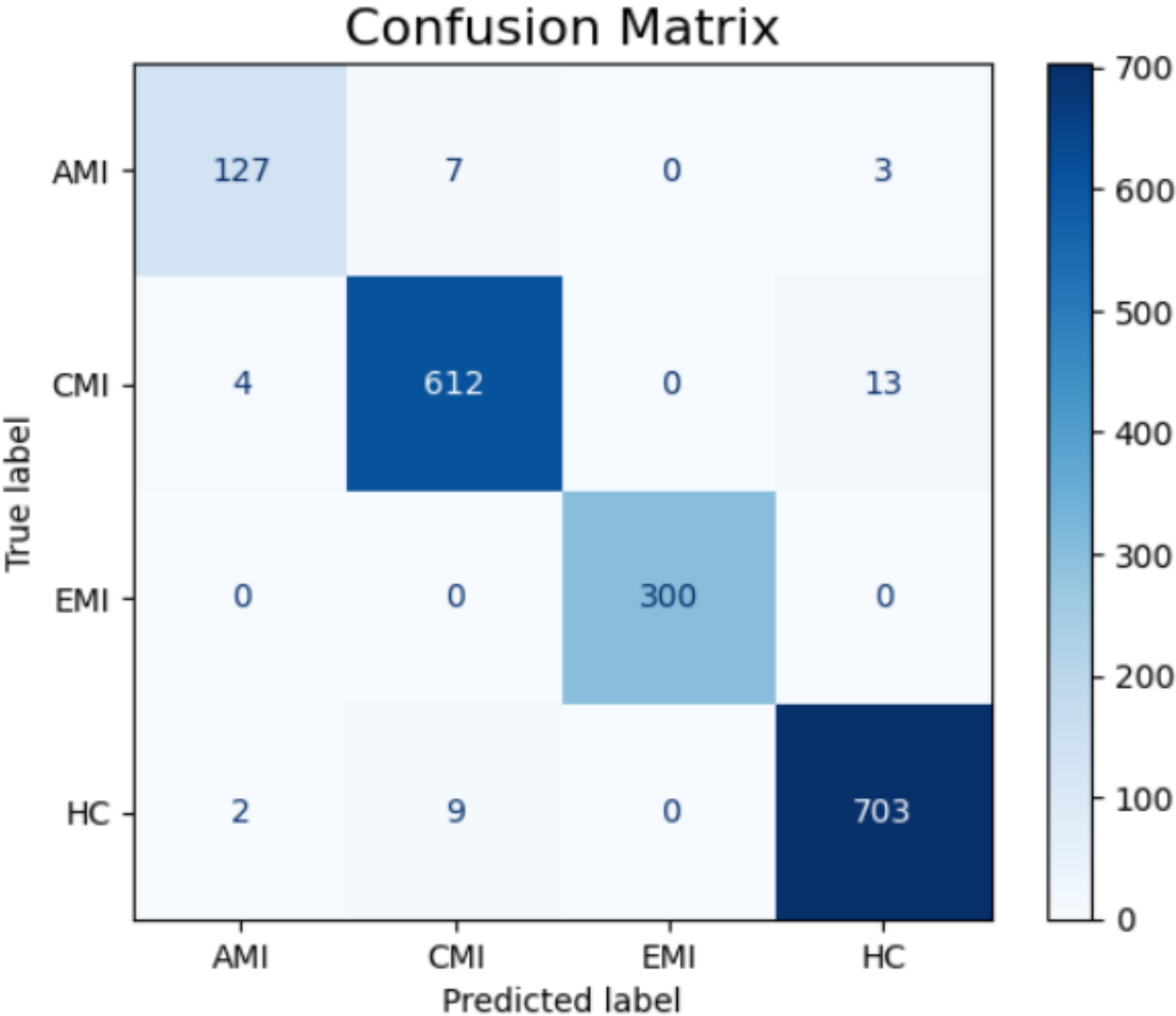
Training vs Validation of CNN-LSTM architecture



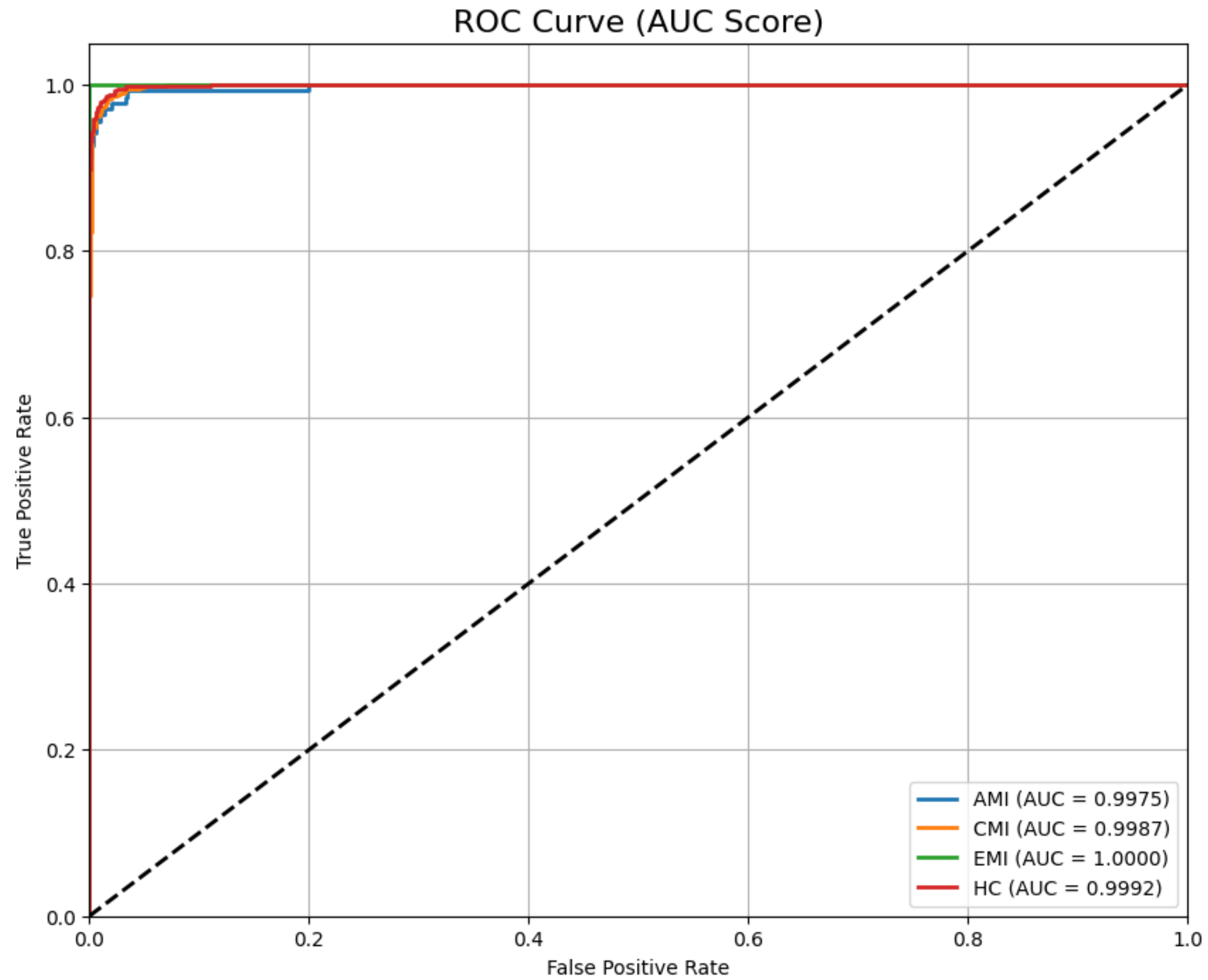
Classification Report of CNN-BiLSTM architecture

Classification Report:

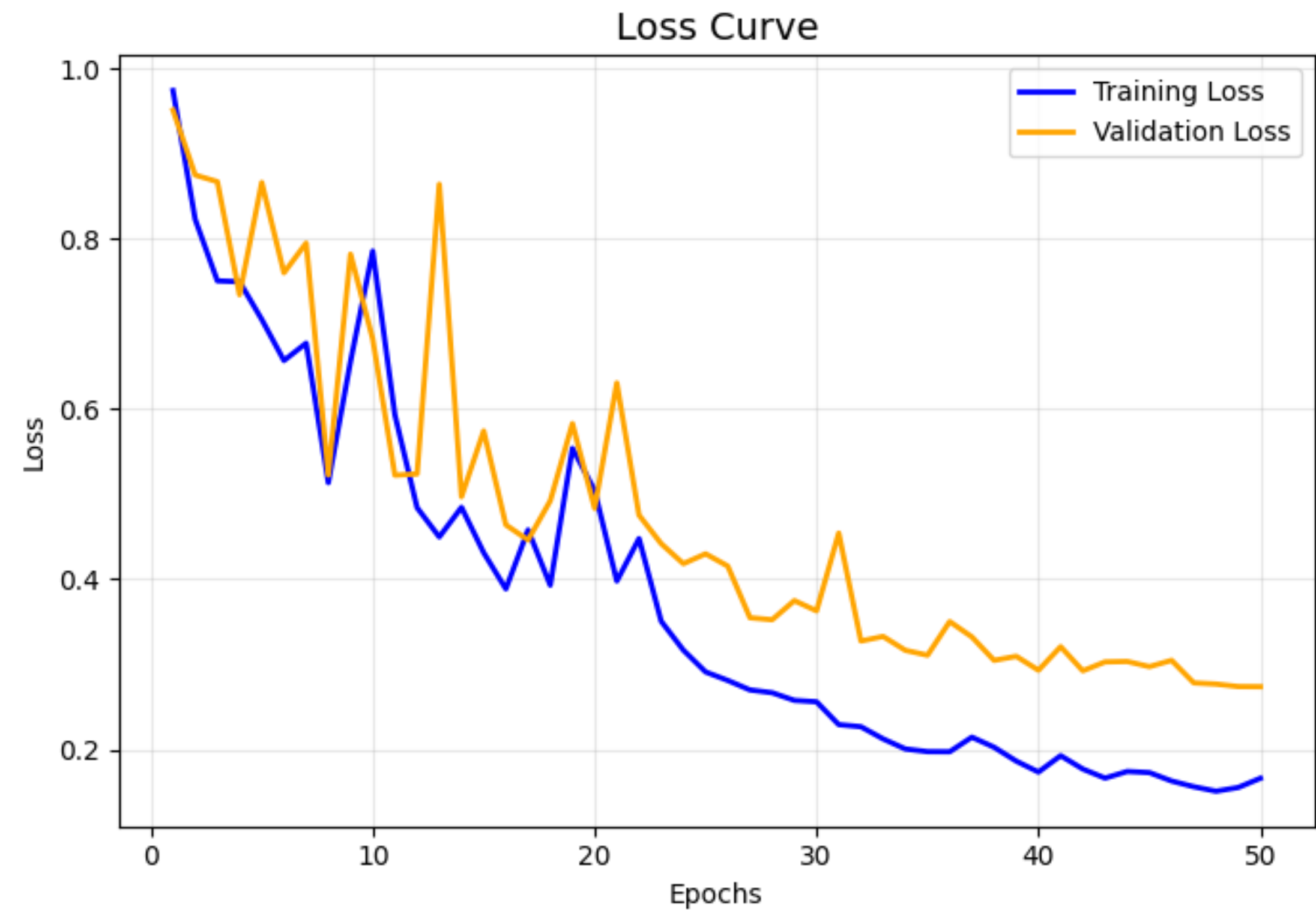
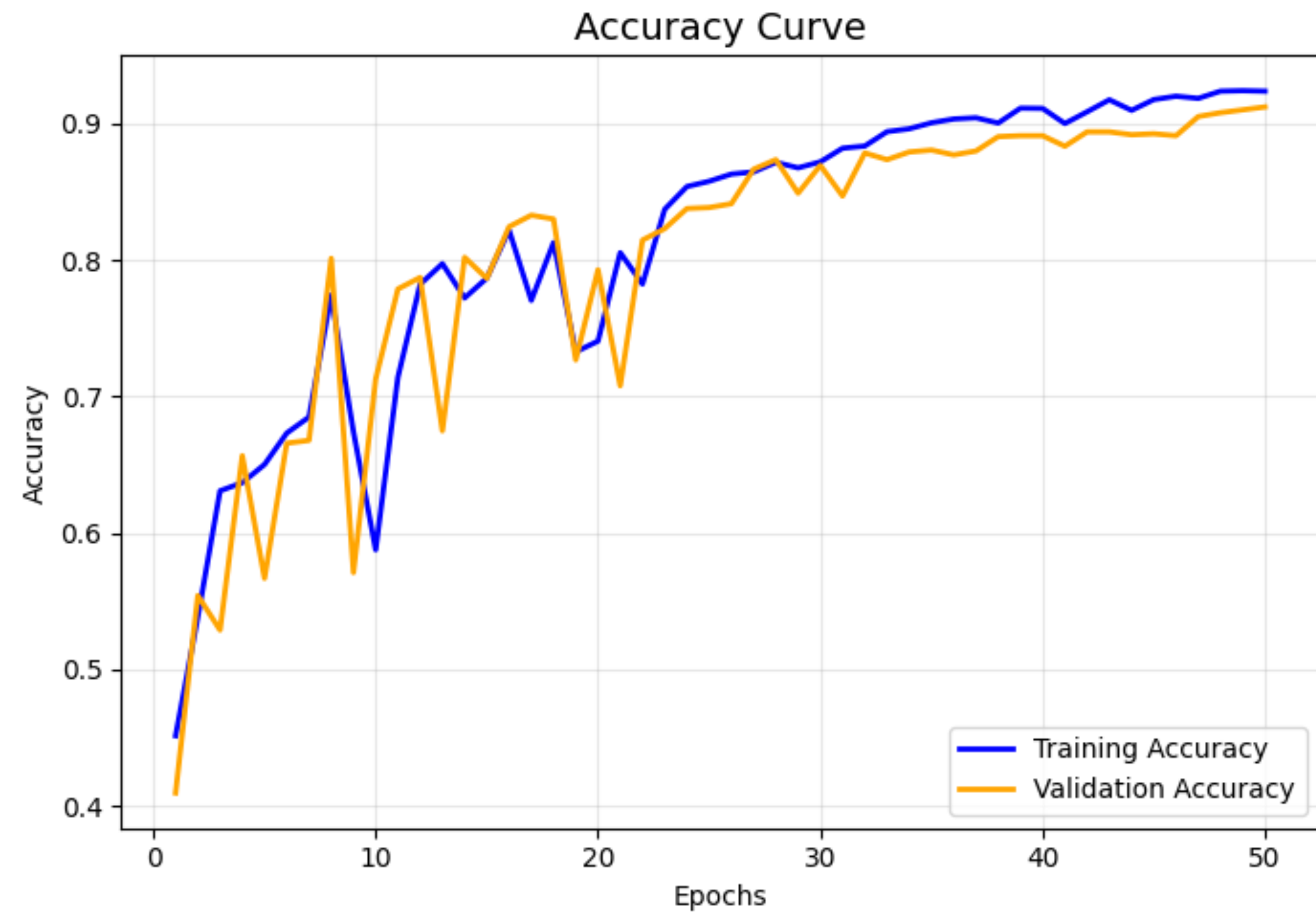
	precision	recall	f1-score	support
AMI	0.95	0.93	0.94	137
CMI	0.97	0.97	0.97	629
EMI	1.00	1.00	1.00	300
HC	0.98	0.98	0.98	714
accuracy			0.98	1780
macro avg	0.98	0.97	0.97	1780
weighted avg	0.98	0.98	0.98	1780



ROC curve of CNN-BiLSTM architecture



Training vs Validation of CNN-BiLSTM architecture

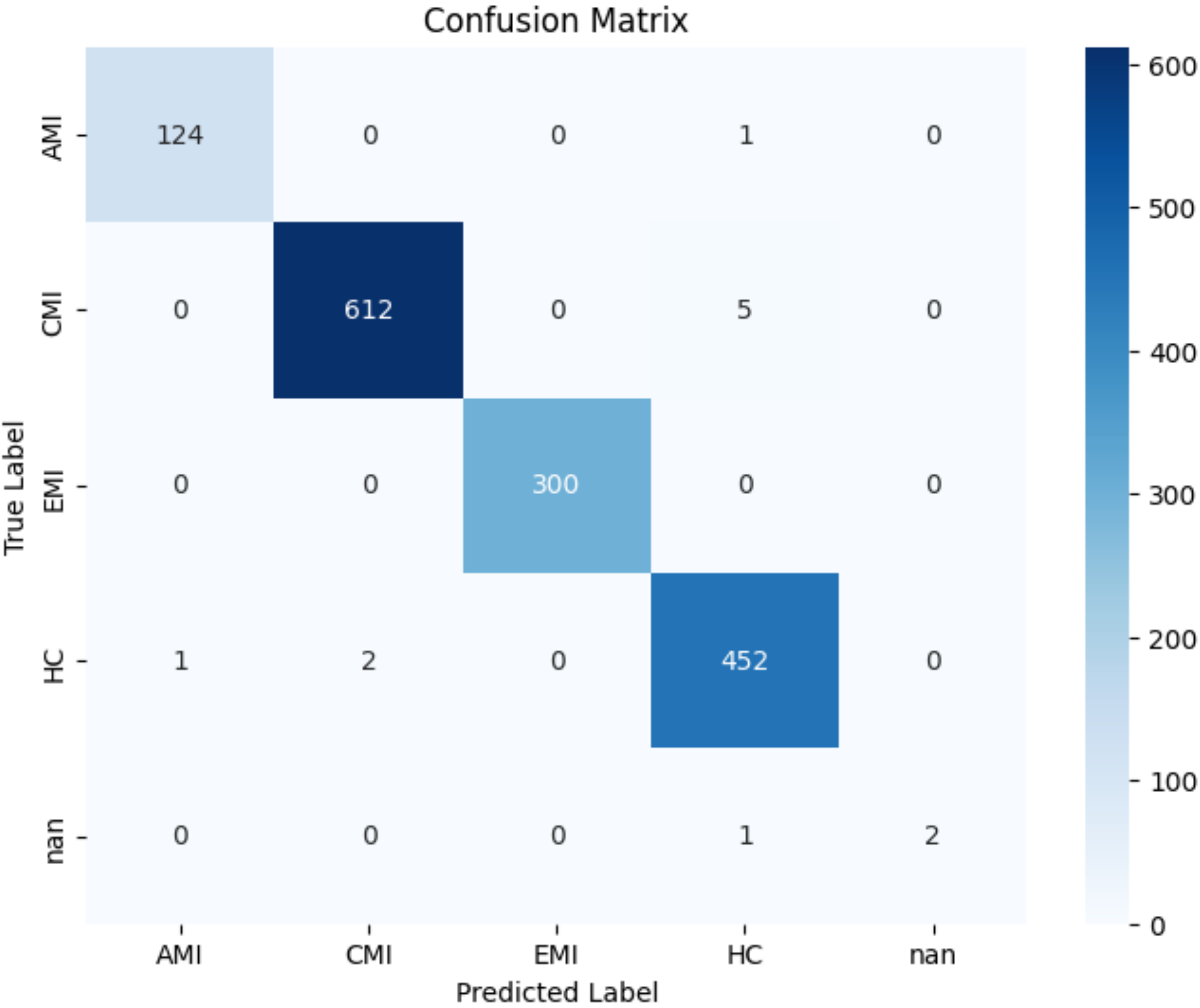


FINAL RESULTS:
Training Accuracy: 92.38%
Validation Accuracy: 91.22%

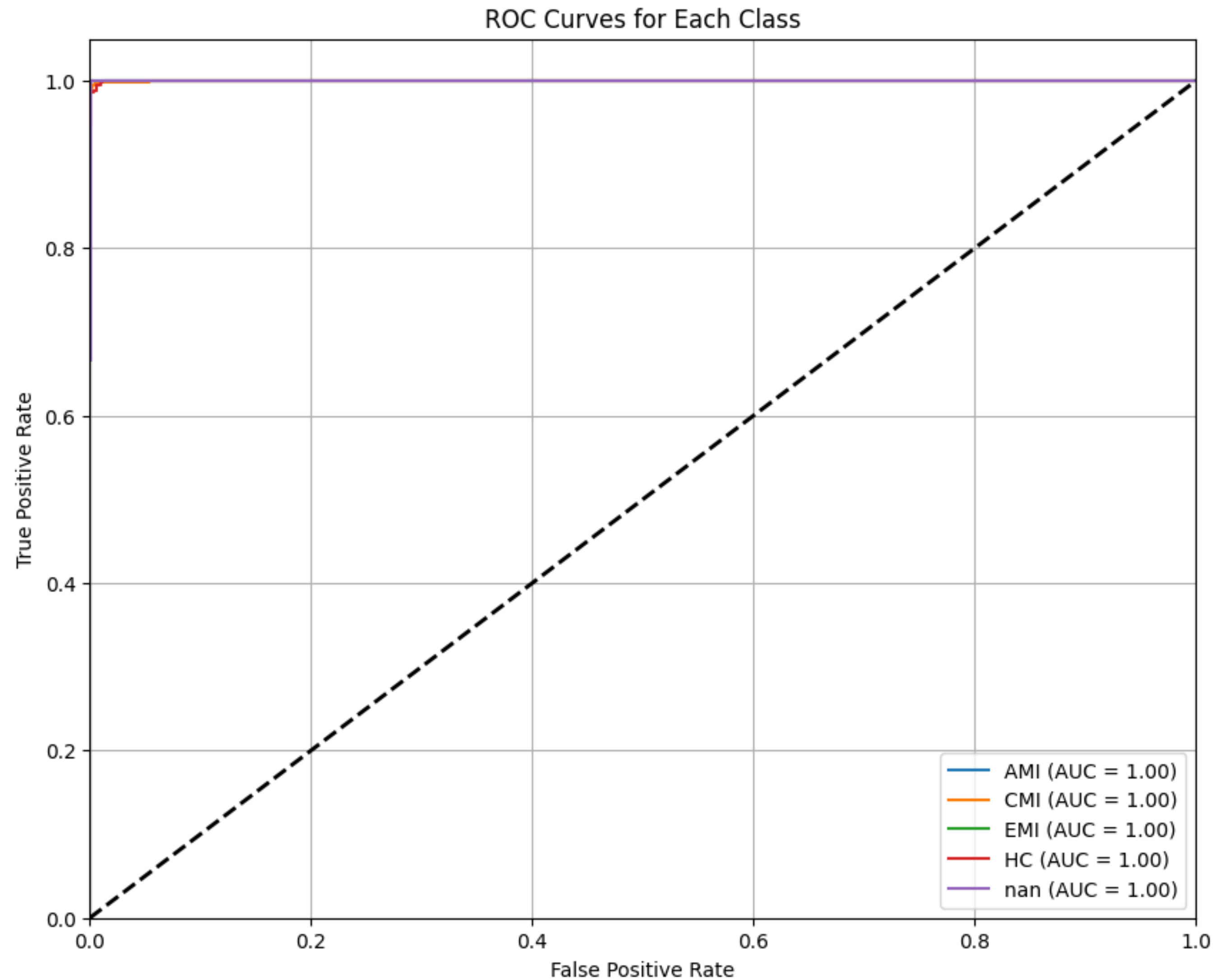
Classification Report of CNN-BiLSTM with Attention mechanism architecture

--- Final Results ---
Test Accuracy: 0.9933
47/47 ————— 2s 47ms/step

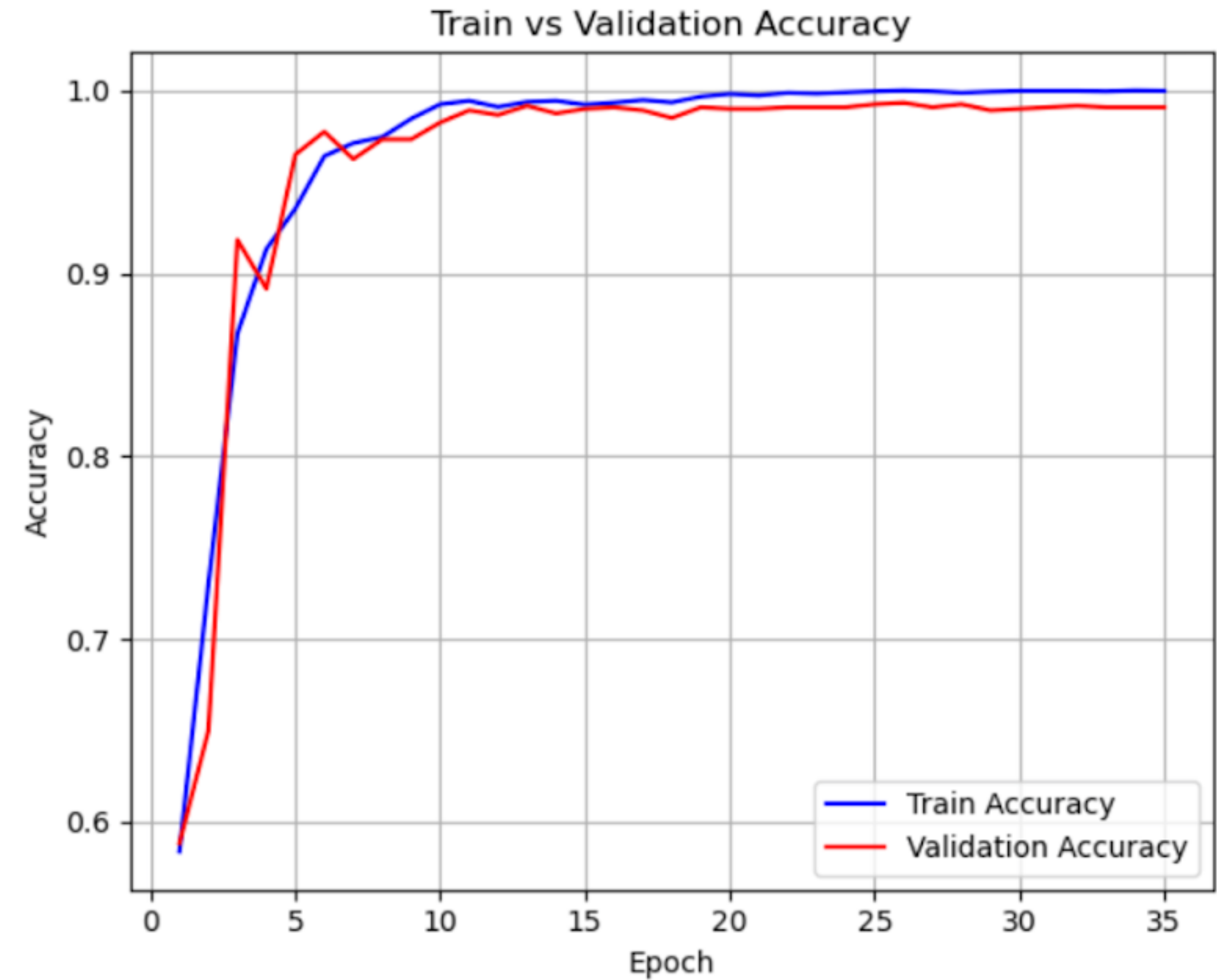
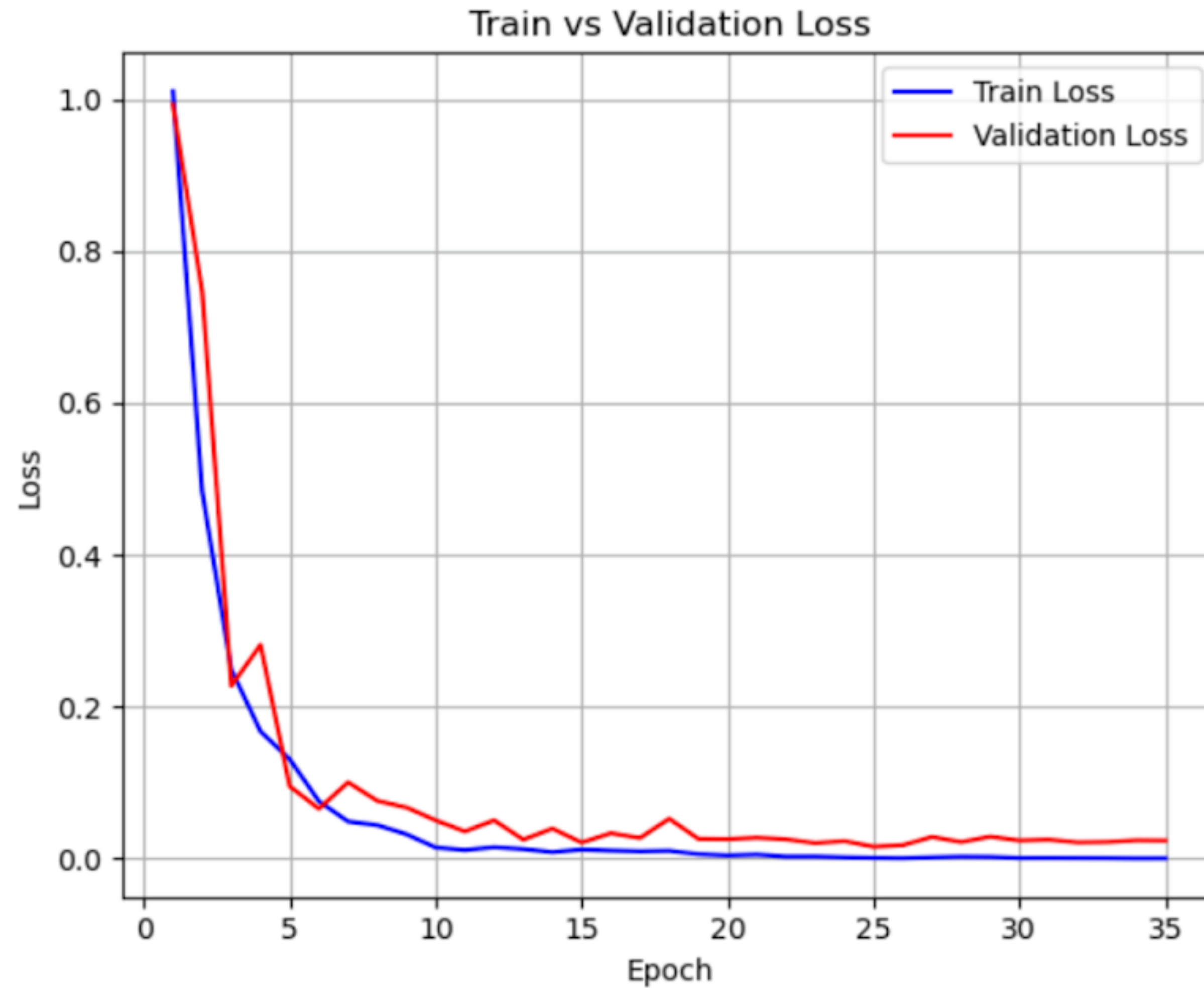
Classification Report:				
	precision	recall	f1-score	support
AMI	0.99	0.99	0.99	125
CMI	1.00	0.99	0.99	617
EMI	1.00	1.00	1.00	300
HC	0.98	0.99	0.99	455
nan	1.00	0.67	0.80	3
accuracy			0.99	1500
macro avg	0.99	0.93	0.96	1500
weighted avg	0.99	0.99	0.99	1500



ROC curve of CNN-BiLSTM with Attention mechanism architecture



Training vs Validation of CNN-BiLSTM with Attention mechanism architecture



Model Parameters

- **Optimizer**

- Type: Adam
- Learning Rate: 0.0001

- **Loss Function**

- Type: Focal Loss
- Gamma: 2.0
- Alpha: 0.25

- **Training Configuration**

- Epochs: 100
- Batch Size: 32
- Validation Split: Patient-based (no leakage)

- **Regularization**

- Dropout Rate: 0.3 (diagnostic) / 0.4 (clinical)
- Batch Normalization: Applied after each dense layer

Real-World Applications and Industry Use

- Useful in hospitals and cardiac care units for rapid heart attack detection and patient monitoring.
- Can support emergency departments and ambulance services in making quick preliminary assessments.
- Applicable in diagnostic centers and ECG labs to improve reporting speed and accuracy.
- Can be integrated into telemedicine platforms for remote cardiac monitoring and rural healthcare support.
- Relevant to medical device and health-tech industries for smart ECG machines and AI-based diagnostic tools.

Future Work and Project Extension

- Improve generalization using multi-hospital and multi-dataset ECG data.
- Reduce bias and strengthen leakage-free patient-level validation.
- Further optimize ST-GNN architecture, feature extraction, and threshold tuning.
- Expand from 4-class prediction to broader cardiac abnormality detection.
- Incorporate explainable AI for clinician-friendly decision support.
- Build a real-time deployment system for wearable and hospital ECG monitoring.

Project Outcomes

Conclusion

- Developed a dual-input ST-GNN for ECG-based myocardial infarction detection.
- Combined raw waveform input with extracted clinical features for improved diagnosis.
- Used leak-proof patient-level splitting and threshold-based evaluation for reliable testing.

Key Achievements

- Built and trained a dual-branch deep learning architecture.
- Integrated 36 ECG-based clinical features with raw signal analysis.
- Established a safer evaluation workflow with leakage audit and confusion-matrix testing.

Final Takeaway

- Hybrid ECG intelligence can improve automated cardiac diagnosis.
- Reliable validation is essential for trustworthy medical AI systems.

Patent Details



Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India



Application Details

APPLICATION NUMBER	202541127318
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	16/12/2025
APPLICANT NAME	1 . Rajagiri School of Engineering and Technology 2 . Ms Anila Kuriakose 3 . Jose Dipu 4 . Jonath Joseph 5 . Joshua Raphel Jolly 6 . Jyothis P
TITLE OF INVENTION	Machine and Deep Learning Powered ECG Monitoring Device for Classification of Myocardial Infarction Stages
FIELD OF INVENTION	BIO-MEDICAL ENGINEERING
E-MAIL (As Per Record)	volksphantomip@gmail.com
ADDITIONAL-EMAIL (As Per Record)	patents@volksphantom.com
E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	--
PUBLICATION DATE (U/S 11A)	02/01/2026

Application Status

APPLICATION STATUS

Awaiting Request for Examination

[View Documents](#)



Filed



Published



RQ Filed



Under Examination



Disposed

Conference Details

- Conference: 2026 IEEE 5th International Conference on Signal Processing, Informatics, Communication and Energy Systems (IEEE SPICES 2026)
- Venue: Government College of Engineering Kannur, Kannur, Kerala, India
- Dates: 09–11 October 2026
- Full paper submission: 15 January 2026 to 15 April 2026
- Acceptance notification: 15 June 2026
- Final camera-ready submission: 15 June 2026 to 30 July 2026
- Registration deadline: 30 July 2026

References

Base Paper

- Prabhakararao, E. and Dandapat, S., 2020. Myocardial infarction severity stages classification from ECG signals using attentional recurrent neural network. *IEEE Sensors Journal*, 20(15), pp.8711-8720.

Supporting Papers

- Alexander, R. Wayne, et al. "Hurst's the heart." *Clinical Cardiology* 23.11 (2009): 872.
- Kumar, Vinay, et al. Robbins and Cotran pathologic basis of disease, professional edition e-book. Elsevier health sciences, 2014.
- Bulluck, Heerajnarain, Derek M. Yellon, and Derek J. Hausenloy. "Reducing myocardial infarct size: challenges and future opportunities." *Heart* 102.5 (2016): 341-348.
- Mellert, Fritz, et al. "Detection of (reversible) myocardial ischemic injury by means of electrical bioimpedance." *IEEE Transactions on Biomedical Engineering* 58.6 (2010): 1511-1518.
- Ansari, Sardar, et al. "A review of automated methods for detection of myocardial ischemia and infarction using electrocardiogram and electronic health records." *IEEE reviews in biomedical engineering* 10 (2017): 264-298.

References

Supporting Papers

- M. Antman et al., “Time is muscle: Translation into practice,” J. Am. College Cardiol., vol. 52, no. 15, pp. 1216–1221, 2008.
- H. Bulluck, D. M. Yellon, and D. J. Hausenloy, “Reducing myocardial infarct size: Challenges and future opportunities,” Heart, vol. 102, no. 5, pp. 341–348, Mar. 2016.
- F. Mellert et al., “Detection of (Reversible) myocardial ischemic injury by means of electrical bioimpedance,” IEEE Trans. Biomed. Eng., vol. 58, no. 6, pp. 1511–1518, Jun. 2011.
- S. Ansari et al., “A review of automated methods for detection of myocardial ischemia and infarction using electrocardiogram and electronic health records,” IEEE Rev. Biomed. Eng., vol. 10, pp. 264–298, Oct. 2017.
- M. James et al., “Physician accuracy in interpreting potential ST-segment elevation myocardial infarction electrocardiograms,” J. Am. Heart Associ., vol. 2, Oct. 2013, Art. no. e000268
- E. Prabhakararao and S. Dandapat, “Automatic quality estimation of 12-lead ECG for remote healthcare monitoring systems,” in Proc. IEEEEMBS Conf. Biomed. Eng. Sci. (IECBES), Dec. 2018, pp. 554–559.

Thank You

Appendix

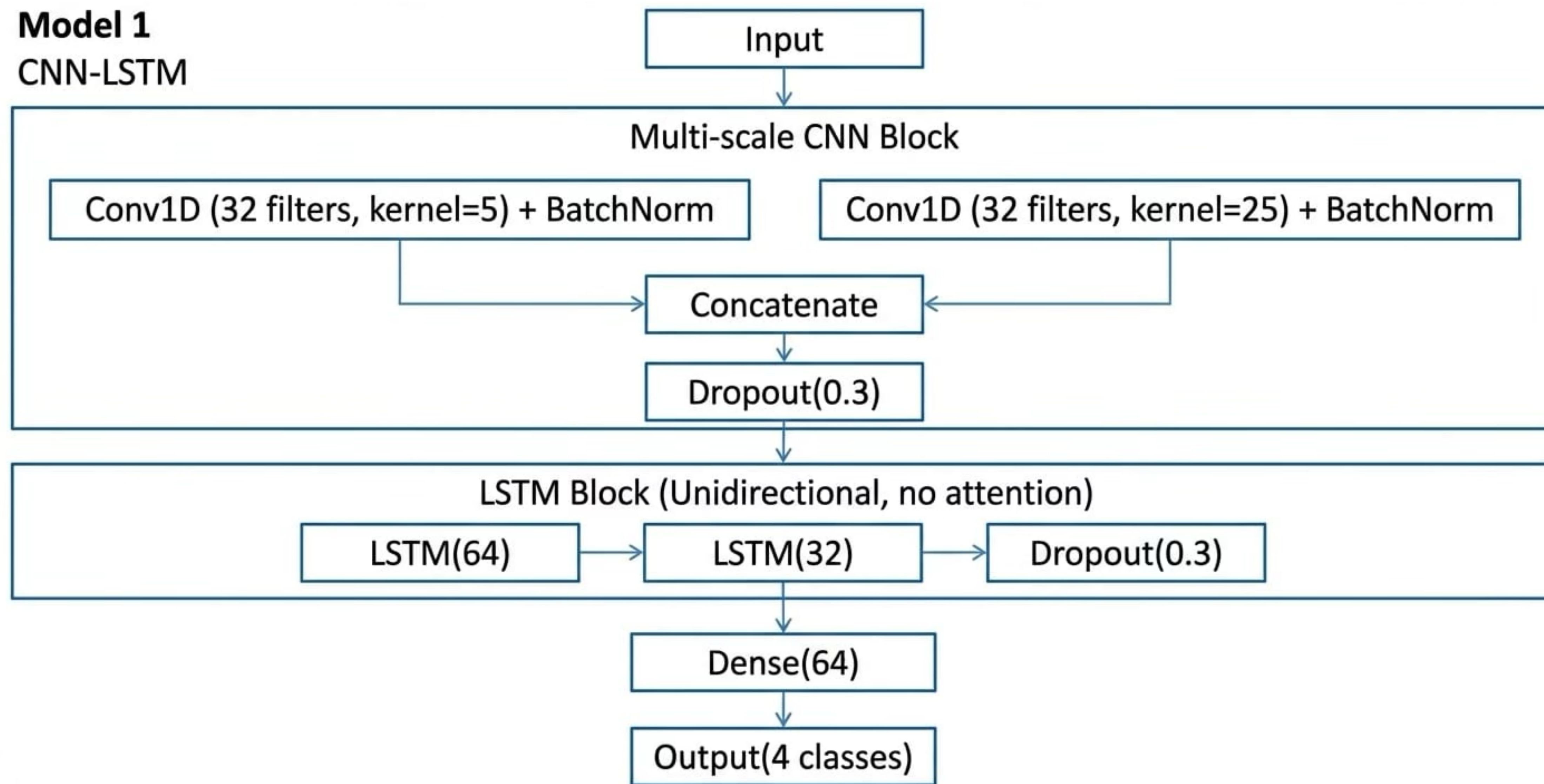
Model 1

CNN-LSTM

- **Architecture** : Input $\rightarrow$ Multi-scale CNN (2 $\times$ Conv1D-32) $\rightarrow$ Single LSTM (64 $\rightarrow$ 32 units) $\rightarrow$ Dense(64) $\rightarrow$ Output(4 classes)
- **CNN Block** : Two parallel Conv1D branches (kernels 5 & 25) with BatchNorm, concatenated + Dropout(0.3)
- **LSTM Block** : LSTM(64) $\rightarrow$ LSTM(32) $\rightarrow$ Dropout(0.3) unidirectional, no attention
- **Total Neurons** : $64 + 32 + 64 + 4 = 164$ neurons

Model 1

CNN-LSTM



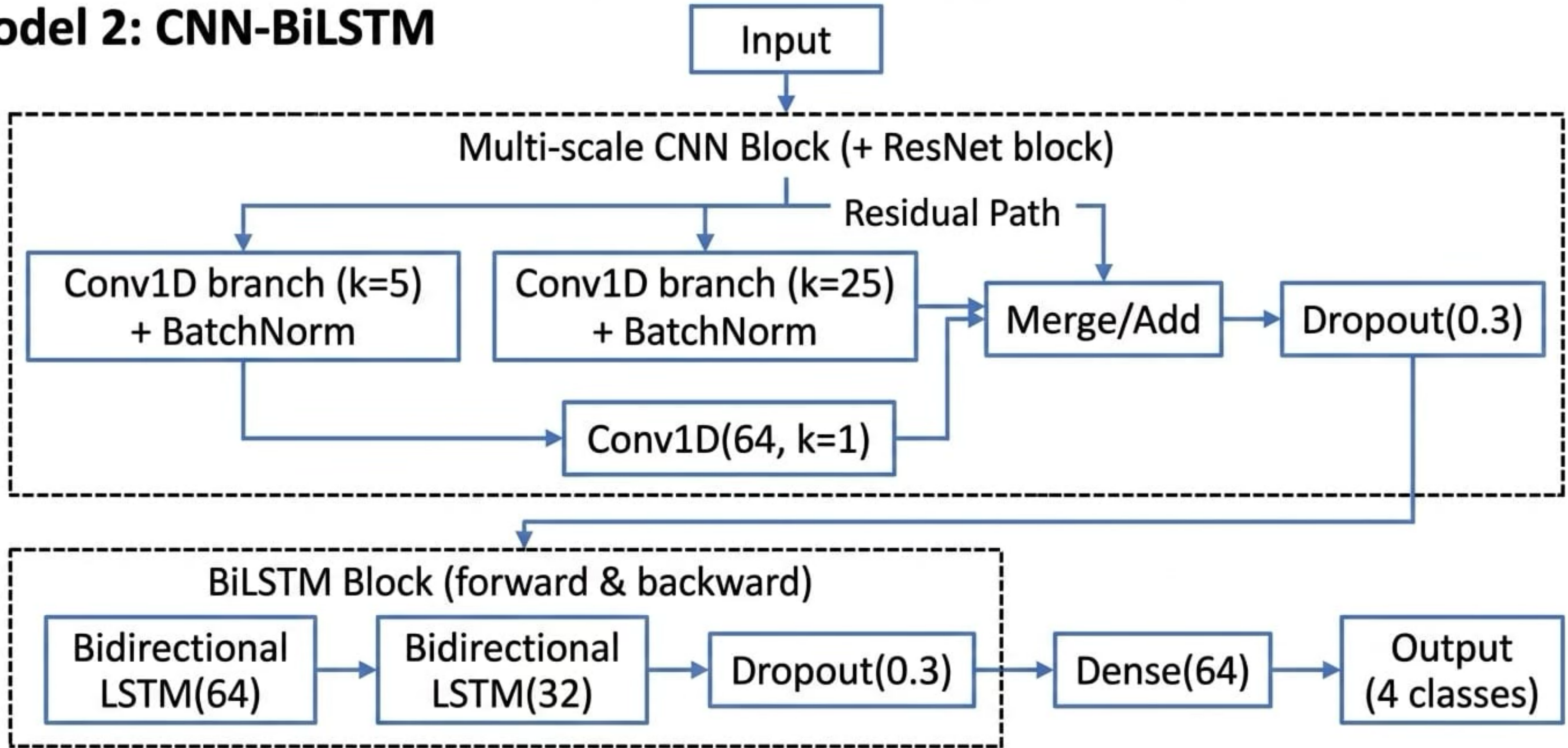
*Total Neurons: $64 + 32 + 64 + 4 = 164$ neurons

Model 2

CNN-BiLSTM

- **Architecture** : Input $\rightarrow$ Multi-scale CNN ($2 \times \text{Conv1D-32} + \text{ResNet block}$) $\rightarrow$ Bidirectional LSTM ($64 \rightarrow 32$ units) $\rightarrow$ Dense(64) $\rightarrow$ Output(4 classes)
- **CNN Block** : Two parallel Conv1D branches (kernels 5 & 25) with BatchNorm, merged + Conv1D(64, kernel=1) residual path, Dropout(0.3)
- **BiLSTM Block** : Bidirectional LSTM(64) $\rightarrow$ Bidirectional LSTM(32) $\rightarrow$ Dropout(0.3) processes forward AND backward in time
- **Total Neurons** : $128 + 64 + 64 + 4 = 260$ neurons

Model 2: CNN-BiLSTM

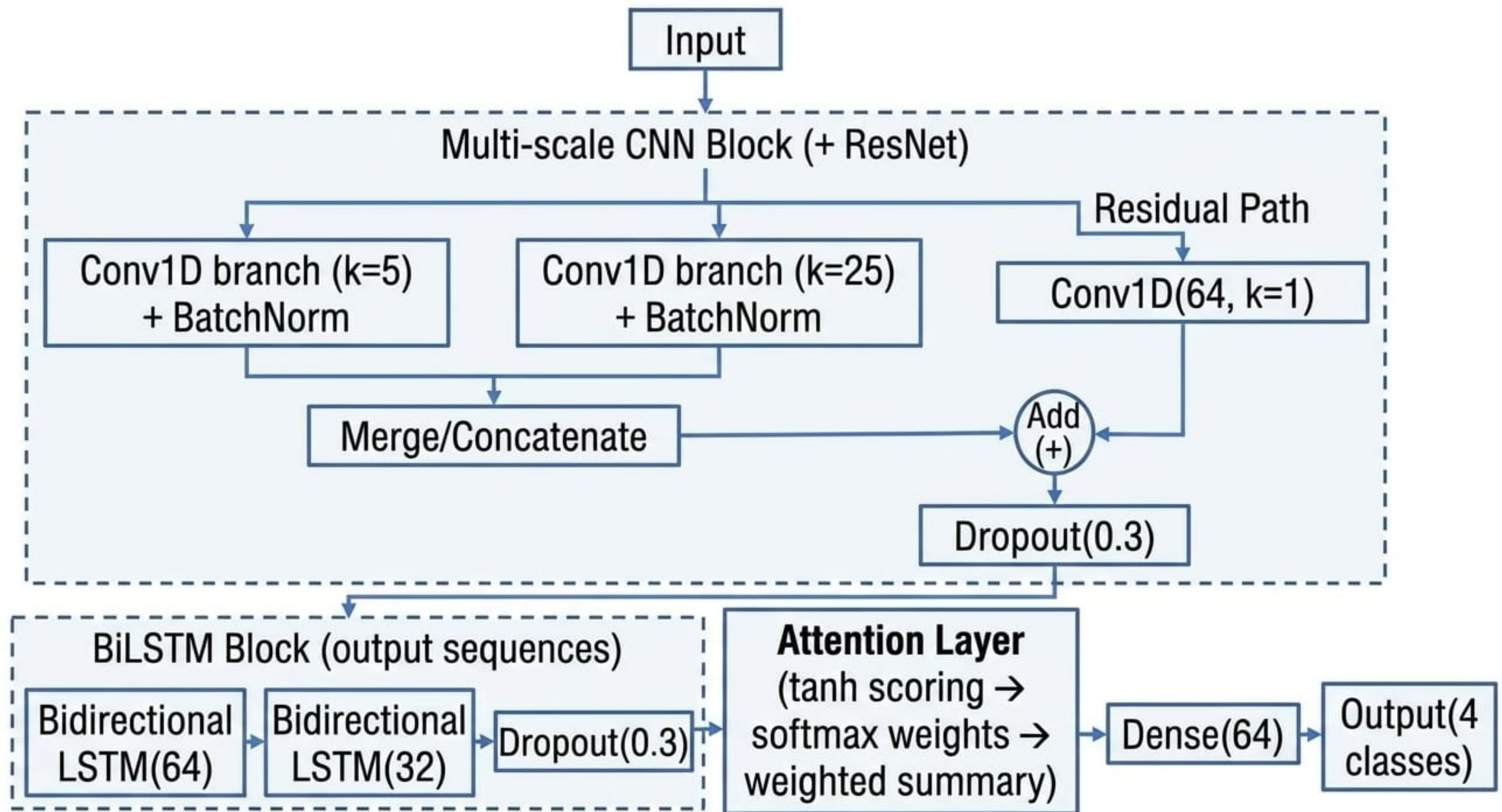


*Total Neurons: $128 + 64 + 64 + 4 = 260$ neurons

Model 3

CNN-BiLSTM with Attention Mechanism

- **Architecture** : Input $\rightarrow$ Multi-scale CNN ($2 \times \text{Conv1D-32} + \text{ResNet block}$) $\rightarrow$ Bidirectional LSTM ($64 \rightarrow 32$ units) $\rightarrow$ Attention Layer $\rightarrow$ Dense(64) $\rightarrow$ Output(4 classes)
- **CNN Block**: Two parallel Conv1D branches (kernels 5 & 25) with BatchNorm, merged + Conv1D(64, kernel=1) residual path, Dropout(0.3)
- **BiLSTM Block**: Bidirectional LSTM(64) $\rightarrow$ Bidirectional LSTM(32) $\rightarrow$ Dropout(0.3) both layers output sequences for attention
- **Attention Layer**: Learns time-step importance weights via tanh scoring, applies softmax, produces weighted summary of LSTM outputs
- **Total Neurons**: $128 + 64 + 64$ (attention scoring) $+ 64 + 4 = 324$ neurons



*Total Neurons: $128 + 64 + 64$ (attention scoring) $+ 64 + 4 = 324$ neurons